Faculty 4: Computer Science

Institute for Web Science and Technologies

Fair-explainable Clustering: from Fairness to Explainability

Thesis Proposal

submitted by

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Zusammenfassung

Abstract

In unsupervised machine learning, clustering algorithms are widely used to discover patterns within complex data. However, standard clustering methods often lack transparency and may exhibit social biases, especially when protected attributes (such as gender or race) are displayed. This thesis aims to systematically evaluate fairness and explainability in clustering by integrating and analyzing existing methods that is both fair—ensuring unbiased group membership with respect to sensitive attributes—and explainable—providing clear, human-understandable rationale for each cluster assignment. The proposed approach integrates fairness constraints directly into the clustering objective, while simultaneously enabling interpretable explanations through rule-based and exemplar-driven methods. Evaluation will be performed on real-world datasets with sensitive attributes, using quantitative fairness and interpretability metrics. The expected outcome is a transparent, equitable clustering tool suitable for responsible deployment in domains with high ethical sensitivity

Contents

1 Introduction

Clustering is an important technique in unsupervised learning because it allows data points to be grouped based on similarity rather than predefined labels. Its uses are many, ranging from customer segmentation to medical diagnostics and social network analysis. Despite their extensive use, traditional clustering methods such as K-means and DBSCAN frequently lack measures for ensuring fairness and transparency [5, 8]. They may unintentionally generate biassed groupings, particularly when dealing with data including sensitive factors such as race, gender, or age.

In recent years, there has been an increased emphasis on the ethical implications of machine learning models, resulting in a greater interest in fairness and explainability [11]. Fairness in clustering aims to prevent biassed outcomes by ensuring that the algorithm's decisions do not disproportionately penalise any one group. Explainability, on the other hand, aims to make the algorithm's decision-making process transparent and understandable to humans.These factors are critical for establishing trust in machine learning systems, especially in high-stakes industries such as healthcare, finance, and criminal justice.

This thesis tackles the combined difficulty of incorporating fairness and explainability into clustering algorithms. The goal is to create systems that provide equitable groupings while also providing explicit, interpretable rationale for their selections.By doing so, the study hopes to close the gap between technical competence and ethical responsibility in unsupervised learning [4, 12].

1.1 Motivation

Clustering is a fundamental technique in unsupervised machine learning that is commonly used to discover hidden structures in data. Traditional clustering algorithms, on the other hand, [19] frequently function as *"black boxes,"* with little transparency and the ability to perpetuate data biases [19]. This opacity presents substantial issues, particularly in high-risk industries such as healthcare, finance, and criminal justice, where understanding the reasoning behind algorithmic decision is critical.

Recent research has demonstrated the growing importance of integrating fairness and explainability into clustering algorithms. Existing analyses of fair clustering approaches highlight limitations in clearly defining utility and addressing downstream consequences in machine learning contexts [11]. Furthermore, the importance of interpretable clustering results has been emphasised to meet ethical and regulatory requirements across various domains [4].

Despite these advancements, there remains a gap in developing clustering methods that are both fair and explainable. This thesis aims to bridge this gap by designing algorithms that ensure equitable treatment of all data subgroups while providing transparent, interpretable results.

1.2 Thesis Objectives

The thesis pursues three key objectives.

First, it aims to conduct a comprehensive literature review of existing approaches to fair and explainable clustering, identifying the current state-of-the-art, their limitations, and open challenges that emerge when fairness and interpretability are combined. This review provides the theoretical foundation for identifying research gaps and motivating the proposed methodology [5,8].

Second, the study will systematically explore different fairness strategies in clustering—such as fairlets, bounded-representation constraints, and proportionality penalties—by evaluating how they influence clustering quality and interact with explainability techniques. This comparative analysis will establish a structured understanding of fairness–utility–interpretability trade-offs [1,3,12].

Finally, the thesis aims to design an evaluation framework for jointly assessing fairness, clustering utility, and interpretability. The framework will employ quantitative metrics and Pareto analysis to capture trade-offs across datasets, providing a reproducible methodology for evaluating clustering models on multiple ethical and performance dimensions [10,18,22].

2 Literature Review

2.1 Fair Clustering

Fair clustering focuses on ensuring that clusters remain unbiased with respect to sensitive or protected group memberships, such as gender, race, or age. Recent research has explored three major strategies to incorporate fairness into clustering algorithms: preprocessing-based methods, fairness-constrained objectives, and fairness-aware adaptations of traditional clustering techniques.

The first category, *preprocessing-based approaches*, ensures fairness by forming balanced micro-groups—commonly referred to as *fairlets*—prior to clustering. Each fairlet satisfies a target group proportion locally, after which standard algorithms such as k-means or k-medoids are applied [7,8]. This approach effectively decouples fairness control from the clustering procedure itself, offering interpretability and compatibility with a range of algorithms. However, fairlet construction can become computationally challenging in high-dimensional data or in the presence of multiple sensitive attributes, and may reduce the flexibility of subsequent clustering stages [3,24].

A second direction integrates fairness constraints directly into the clustering objective. These *in-objective approaches* modify traditional objectives—such as minimizing within-cluster variance—to include fairness regularizers or proportionality penalties that bound group representation within clusters [5,7]. This method provides strong theoretical guarantees and direct control over fairness–utility trade-offs through tunable parameters. However, such formulations often lead to complex optimization problems that require approximations or relaxations to remain computationally feasible.

The third strategy extends fairness principles to more complex clustering paradigms, such as spectral or hierarchical clustering [13]. These adaptations embed fairness constraints within the similarity representation or hierarchical structure, enabling fairness-aware analysis of multi-level or networked data. Across all these methods, fairness is typically assessed using cluster-level metrics—such as group balance or bounded-representation deviation—rather than classification-oriented measures. Collectively, these approaches illustrate a shift toward formalizing fairness in unsupervised settings, balancing interpretability, computational tractability, and fairness guarantees [1,7].

2.2 Explainable Clustering

Explainable clustering extends the principles of Explainable Artificial Intelligence (XAI) to unsupervised learning, where no ground-truth labels exist. The central challenge lies in making cluster assignments interpretable—i.e., understanding why a particular instance belongs to a specific cluster based solely on intrinsic data patterns. Existing work has primarily converged on three complementary explanation paradigms: exemplar-based, rule-based, and feature-attribution approaches [2].

Exemplar-based explanations represent each cluster through a compact set of representative or boundary examples that summarize its essential characteristics. These exemplars enhance interpretability by providing tangible, human-understandable illustrations of cluster structure and composition.A convex clustering framework that integrates exemplar-based models directly into the clustering process, optimizing exemplar selection to capture both representational coverage and cohesion [15]. This approach laid the foundation for later works that formalize exemplar selection as an optimization problem balancing interpretability and cluster fidelity. While such methods improve transparency, they can be sensitive to feature scaling and may yield unrepresentative exemplars in heterogeneous datasets, motivating further research on robust prototype selection strategies for complex or high-dimensional data.

Rule-based explanations generate human-readable if–then statements or decision-tree paths describing cluster membership conditions [20–22]. These rules enable audits of model behavior and can explicitly exclude sensitive features to promote fairness. The primary trade-offs involve balancing rule simplicity (fewer, shorter rules) with fidelity (agreement between rule predictions and actual cluster assignments). Additionally, stability analysis across data resamples has become a critical component in evaluating the reliability of rule-based explanations.

Feature-attribution methods quantify the contribution of individual features to cluster assignments, often adapting techniques originally designed for supervised models, such as SHAP or LIME [18,19]. In clustering, attributions are derived through comparisons between centroids or medoids, or through local density approximations, to identify which attributes drive similarity within clusters. These methods facilitate the detection of spurious correlations or proxy variables that could compromise interpretability or fairness.

Across all explainability approaches, two central criteria emerge: *faithfulness*—ensuring explanations accurately reflect the true clustering behavior—and *usability*—ensuring explanations remain concise and understandable for human decision-makers. Building on these insights, this thesis integrates exemplar- and rule-based explanations with feature-attribution stability analyses to develop a practical, multi-view explainability framework for fair clustering.

2.3 Research Gap

Although fairness and explainability in clustering have been studied extensively, they have evolved largely in isolation. Only limited work exists that systematically evaluates the integration of both fairness and explainability within a unified framework that provides human-understandable reasoning for cluster assignments [7,17].

Most existing fairness mechanisms—such as fairlet preprocessing, bounded-representation constraints, or proportionality penalties—focus on ensuring balanced group composition but typically stop at aggregate fairness metrics. These methods rarely provide cluster-level or instance-level explanations that justify why specific assignments are fair [5,7,8].

Conversely, explainable clustering methods, including exemplar-, rule-, and attribution-based techniques, enhance interpretability but are often applied after unconstrained clustering. As a result, they lack fairness guarantees by design.

Finally, there is still no standardized framework for jointly evaluating fairness, clustering quality, and interpretability. This absence of a unified evaluation approach hinders practical deployment in real-world, high-stakes applications [7,17].

2.4 Research Questions

RQ1: What is the current state of research in fair and explainable clustering, and what are the key challenges when integrating fairness with interpretability?

RQ2: How do different fairness strategies (e.g., fairlets, bounded-representation, proportionality constraints) affect clustering outcomes, and how do they interact with explainability methods such as exemplars, rule-based models, and attribution techniques?

RQ3: What are the practical implications of deploying fair and explainable clustering algorithms in high-stakes domains?

3 Methodology

3.1 Methods for Fair and Explainable Clustering

This thesis will evaluate the integration of fairness mechanisms with explanation modules within a unified pipeline. The focus is on analyzing trade-offs rather than designing new algorithms.

The fairness component will employ two representative strategies. The first, fairlet-style preprocessing, constructs small balanced micro-clusters (fairlets) that locally satisfy target group proportions before applying a standard clustering algorithm such as k-means or k-medoids [7,8]. By controlling fairlet size and group ratios, this approach ensures local fairness while maintaining compatibility with conventional clustering pipelines.

The second approach, the bounded-representation objective, modifies the clustering formulation to constrain group proportions per cluster within predefined limits or penalize deviations from target distributions [1, 5, 7]. Adjusting these constraint weights allows for systematic exploration of fairness–utility–explainability trade-offs.

To enhance interpretability, the explanation module will combine exemplar-based, rule-based, and feature-attribution methods. Exemplar-based explanations summarize cluster characteristics by identifying representative points such as medoids or prototypes. Rule-based explanations generate human-understandable conditions—learned through decision trees or global rule models—that describe membership criteria while excluding sensitive attributes [2, 21]. Finally, feature attribution methods (e.g., SHAP-style analyses adapted for clustering) provide quantitative insight into which features most influence cluster membership, enabling robustness and stability checks [2,19].

Together, these modules form an integrated workflow, where the fairness mechanism determines the cluster assignments, and the explainability stage provides faithful interpretations of these assignments.

Integration details: In the evaluation pipeline, the fairness stage (fairlets or bounded-representation) determines cluster assignments; the explanation stage then operates on these assignments. The analysis will verify whether explanations remain faithful to fairness-shaped decisions [1,2].

3.2 Datasets and Preprocessing

The experiments will be conducted on widely used fairness-aware benchmark datasets, including *Adult*, *COMPAS*, *Bank Marketing*, *German Credit*, and *Diabetes* [16]. These datasets contain sensitive attributes such as gender, race, or age, which are explicitly isolated for fairness constraints but excluded from model inputs to avoid bias leakage. This selection ensures a representative mix of socio-economic, demographic,

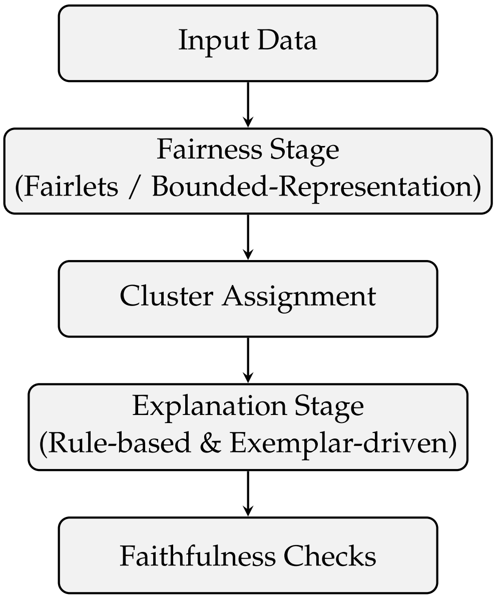

Figure 1: Workflow of fairness-aware explainable clustering. Explanations are grounded in fairness-constrained cluster assignments, with additional faithfulness checks.

and health-related domains where fairness and interpretability are critical for ethical decision-making.

All datasets will undergo a standardized preprocessing pipeline involving missing value handling, categorical encoding, and feature scaling. Variance-based feature selection and correlation analysis will be applied to remove redundant features and maintain consistent feature representations across datasets. Each preprocessing step will be implemented with fixed random seeds and stratified sampling to ensure reproducibility and stable fairness assessment across resamples.

Feature Selection. For high-dimensional datasets, variance thresholding will be applied to remove uninformative or low-variance features that do not contribute to cluster differentiation. When features exhibit strong correlations, domain-specific dimensionality reduction techniques—such as Principal Component Analysis (PCA) or correlation-based pruning—will be considered to minimize redundancy while preserving interpretability. Sensitive attributes (e.g., gender, race, or age) will be carefully excluded from the feature selection process to prevent fairness leakage. All feature selection methods will be consistently applied across datasets to ensure comparability of results and to maintain a uniform experimental design.

Preprocessing Pipeline. The preprocessing pipeline will include several standardized steps to prepare data for fairness-aware clustering. Missing values will be imputed or removed depending on their sparsity and relevance to model performance. Categorical variables will be encoded using one-hot encoding for nominal features or ordinal encoding when appropriate. Continuous features will be normalized using standardization or min–max scaling to maintain consistent distance metrics for clustering algorithms. Sensitive attributes will be isolated for fairness analysis and reporting but will not be used as input features during clustering. Additional steps such as outlier filtering or de-duplication may be applied when justified by the dataset documentation or exploratory analysis.

Data Splits and Reproducibility. For evaluation, each dataset will be partitioned using repeated stratified resampling based on sensitive attributes (for example, five random splits) to examine the stability of fairness and explanation metrics. Random seeds will be fixed across all runs to ensure consistent initialization and reproducibility of clustering outcomes. Logging and configuration files will be maintained for each experimental setup, enabling transparent tracking of parameters and reproducible experiment reruns.

3.3 Baselines

To comprehensively evaluate the proposed fairness-aware and explainable clustering framework, this study will benchmark its performance against both conventional and state-of-the-art baselines. These baseline models serve as reference points for assessing the effectiveness and trade-offs introduced by incorporating fairness and interpretability mechanisms into clustering algorithms.

The non-fair baselines will include traditional clustering methods such as k-means, spectral clustering, and DBSCAN. These algorithms are widely adopted in unsupervised learning and provide a foundation for measuring how fairness constraints and explainability modules influence clustering performance. Their results will help quantify the degree to which the proposed fairness mechanisms modify cluster structure, separability, and overall utility compared to unconstrained approaches.

For fairness-aware baselines, two representative strategies will be implemented. The first employs a *fairlet-based preprocessing pipeline*, in which data instances are grouped into locally balanced micro-clusters (“fairlets”) that respect target group proportions prior to standard clustering [8]. The second baseline adopts a *bounded-representation objective*, which embeds fairness constraints or proportionality penalties directly within the clustering optimization process [1]. These two approaches are complementary: the fairlet method enforces fairness at the data preprocessing stage, while the bounded-representation formulation integrates fairness directly into the model objective. Comparing these strategies will allow for a systematic investigation of their relative performance, fairness guarantees, and computational trade-offs.

To evaluate explainability, several explanatory configurations will be analyzed: exemplar-based explanations, rule-based explanations, and hybrid models that combine both approaches, optionally enhanced with feature attribution overlays. The exemplar-only configuration highlights representative data points to intuitively summarize each cluster, whereas rule-based models produce explicit, human-readable rules that describe cluster membership boundaries. The hybrid configuration seeks to combine the interpretive strengths of both techniques, maximizing transparency and faithfulness. Additionally, rule complexity constraints (e.g., limiting decision-tree depth or rule count) will be explored to examine the trade-off between interpretability and fidelity.

3.4 Hyperparameters and Model Selection

Hyperparameter tuning and model selection will be performed systematically across the fairness, clustering, and explainability components to ensure reproducible and well-balanced results. Each component introduces parameters that significantly influence algorithmic behavior, trade-offs, and evaluation outcomes.

For the fairness mechanisms, parameters such as the fairlet size and group ratio targets (in fairlet-based clustering) will be varied to control the level of local balance, whereas proportionality bounds or penalty weights (in bounded-representation models) will be adjusted to enforce fairness at the global clustering level. These hyperparameters determine the strength of fairness enforcement and consequently affect both cluster compactness and representational quality.

In the clustering component, the number of clusters (k) will be determined using internal validation criteria such as the Elbow method, the Silhouette coefficient, and the Davies–Bouldin index, supported by domain-specific knowledge where applicable [9, 14, 23]. The Elbow method provides an intuitive visual approach by identifying the point at which adding more clusters yields diminishing returns in reducing within-cluster variance, while the Silhouette and Davies–Bouldin indices quantitatively assess cohesion and separation between clusters. Additional parameters, including initialization schemes, kernel width for spectral clustering, and convergence criteria, will be systematically tuned to identify stable and high-performing configurations across datasets

For the explainability component, hyperparameters such as the number of exemplars per cluster, the maximum depth of rule-based decision trees, and the neighborhood size for feature attribution will be optimized to balance clarity and fidelity. The inherent trade-off between simplicity and interpretability will be explored through constraint-based tuning, prioritizing parsimonious models that maintain comparable explanatory performance to more complex alternatives.

Model selection will follow a multi-stage protocol. Initially, a grid search will be conducted across the combined parameter spaces of fairness, clustering, and explainability modules. Each configuration will be evaluated on multiple resampled datasets using the metrics defined in Section 3.5. Configurations that fail to meet minimum fairness thresholds (e.g., proportionality deviation exceeding $\delta > 0.1$) will be excluded regardless of clustering performance. Among the feasible configurations, models located on the *Pareto frontier*—representing optimal trade-offs between fairness, clustering utility, and interpretability—will be prioritized for deeper analysis [10, 26]. Three representative models will then be reported: one optimized for fairness, one for clustering utility, and one reflecting a balanced compromise across all objectives. This multi-objective selection approach ensures an equitable and transparent evaluation framework while emphasizing reproducibility and robustness.

3.5 Evaluation Metrics

To rigorously assess the proposed fairness-aware and explainable clustering framework, multiple quantitative metrics will be employed to evaluate clustering utility, fairness, and interpretability. Together, these measures provide a holistic understanding of algorithmic performance and the trade-offs between competing objectives.

Clustering Quality (Utility). The quality of clustering outcomes will be quantified using standard internal validation indices. The Silhouette coefficient [23] measures the cohesion and separation of clusters, indicating how well individual samples fit within their assigned groups. The Davies–Bouldin index [9] serves as a complementary indicator, capturing intra-cluster compactness and inter-cluster distinctness, with lower values reflecting superior cluster formation. In addition, inertia (within-cluster dispersion) [25] will be analyzed to evaluate the overall tightness and homogeneity of clusters. Collectively, these utility metrics provide a robust quantitative baseline for assessing the impact of fairness and interpretability constraints on clustering structure.

Fairness Evaluation. Fairness will be assessed using established cluster-level fairness notions introduced in the literature. The first metric, *Balance* [8], measures the ratio of minority to majority group representation within each cluster and serves as a foundational indicator of group fairness in clustering. To capture stricter compositional fairness, the *Bounded-Representation* criterion [1] constrains each cluster's group proportions to lie within pre-specified lower and upper bounds, quantifying the proportion of clusters satisfying these bounds. Finally, the *Proportional Fairness* [6] ensures that no sufficiently large subset of data points can all be reassigned to a single cluster in a way that would reduce their distance cost, thereby preventing "blocking coalitions." Each metric will be reported per dataset, accompanied by per-cluster group composition tables to ensure transparency and interpretability.

Explainability Evaluation. Explainability will be assessed through a combination of quantitative and qualitative indicators designed to evaluate clarity, simplicity, and stability.*Exemplar clarity and coverage* will quantify how effectively representative samples summarize each cluster, measured by the proportion of data points within a defined proximity of exemplars and by the diversity of exemplars across clusters [15]. This approach follows exemplar-based clustering and prototype selection methods that emphasize interpretability through representative real instances . *Rule simplicity and fidelity* will be examined using metrics such as the number of extracted rules, average rule depth, and fidelity of rule-based predictions with respect to true cluster assignments on held-out resamples [2]. To evaluate robustness, *attribution stability* will be analyzed via correlation and rank consistency of feature importance scores across multiple resamples, supplemented with sensitivity analyses to detect biased or proxy-dependent explanations [19]. These metrics collectively ensure that the generated explanations are interpretable, consistent, and faithful to the underlying clustering process.

Finally, a multi-objective evaluation will be performed through **Pareto front visualizations** [10, 26], jointly analyzing trade-offs between fairness, clustering utility, and interpretability. This integrated assessment framework enables transparent comparison of models and provides insights into how fairness interventions influence both performance and explainability in practical deployment scenarios.

Table 1: Thesis Timeline: Explainable Fair Clustering (Oct 1 – March 1, 2026)

Phase / Month	**Key Tasks and Milestones**
Phase 1: Month 1 (Literature Review & Planning)	- Conduct focused literature review on fair clustering (fairlets, bounded-representation, spectral) and explainable clustering (exemplars, rules, attributions) - Finalize thesis problem statement and unified objective (fair + explainable) - Refine research questions and evaluation criteria (fairness, utility, interpretability)
Phase 2: Month 2 (Method Design)	- Specify unified method: select primary fairness mechanism (fairlets or bounded-representation) and explanation suite - Define constraints/targets (group ratios/bounds/penalties) and explanation constraints (exclude sensitive attributes) - Prepare datasets (Adult, COMPAS, Bank Marketing; optional German Credit) and preprocessing pipeline
Phase 3: Month 3 (Implementation & Initial Testing)	- Implement fairlets or bounded-representation objective; integrate clustering backend (e.g., k-means/medoids) - Implement explanation modules (exemplars, rules, attribution) - Run smoke tests on one dataset; debug pipeline and logging
Phase 4: Month 4 (Evaluation & Analysis)	- Full experiments across datasets with repeated resampling - Hyperparameter sweeps; collect fairness, utility, and interpretability metrics - Statistical testing and Pareto trade-off analysis
Phase 5: Month 5 (Results & Writing)	- Compile quantitative and qualitative results (tables, figures, case studies) - Draft Methodology, Results, and Discussion sections - Internal review and iteration (advisor/peers)
Phase 6: Month 6 (Revision & Finalization)	- Address feedback; finalize text and figures - Polish LaTeX, references, and appendices; ensure reproducibility pack (code + HOWTO) - Submit final thesis

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